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DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

APPLIED ELECTROMAGNETICS @POLIMI

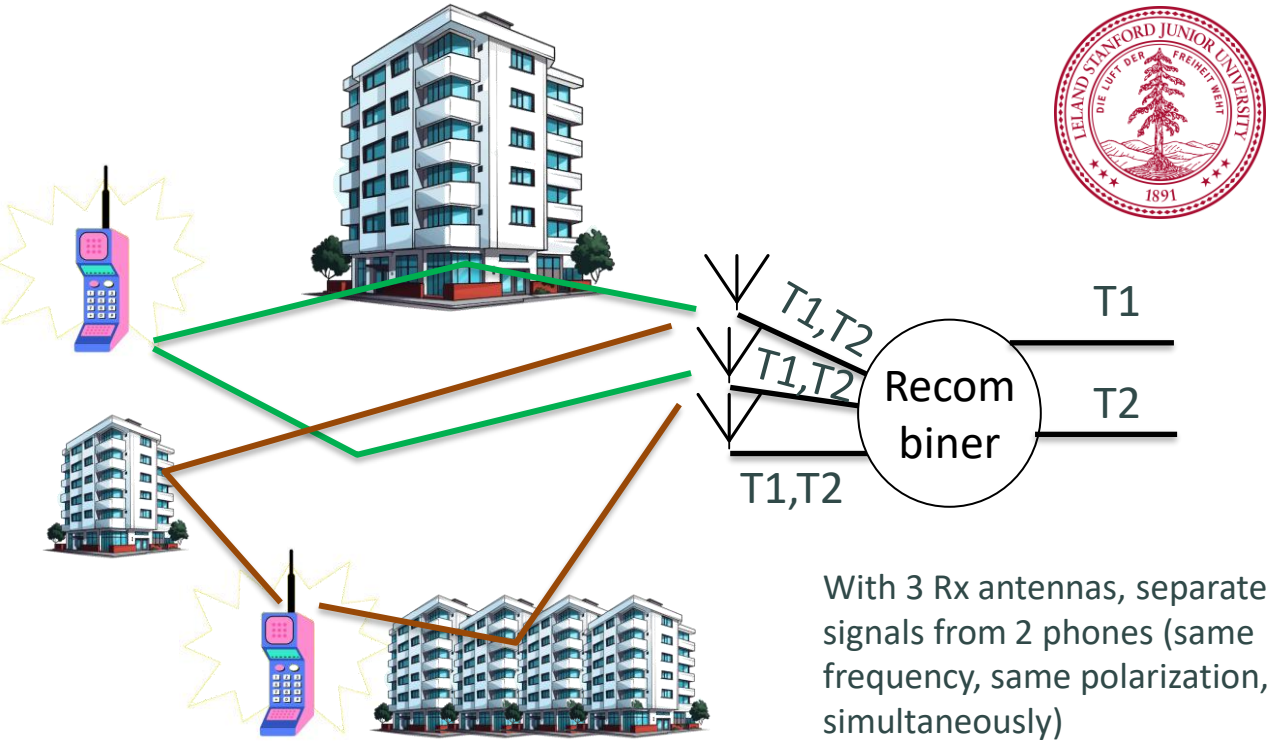
RF Systems

Multi-antenna System Architectures

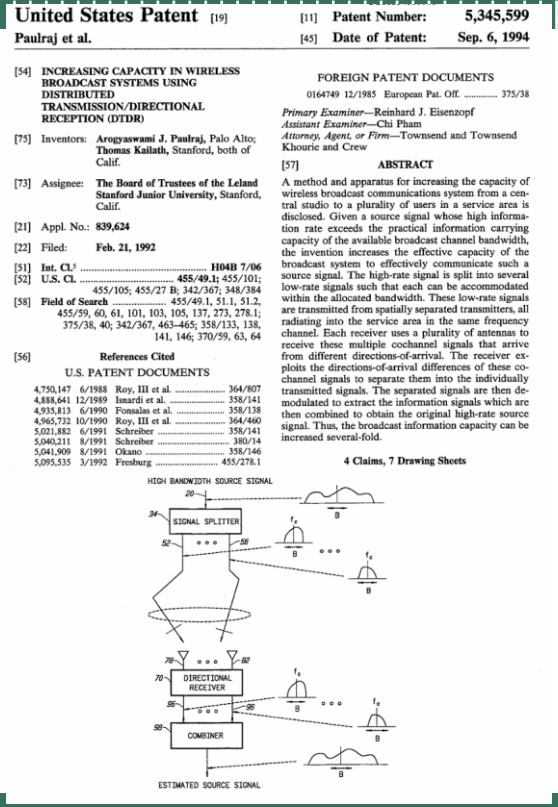
Agenda

- MIMO
- Phased Arrays
- Multi-carrier
- Wrap-up: Architectures and Components

MIMO: Patented in 1991 by Paulraj, Kailath



With 3 Rx antennas, separate signals from 2 phones (same frequency, same polarization, simultaneously)



United States Patent [19]

Paulraj et al.

[11] Patent Number: 5,345,599

[45] Date of Patent: Sep. 6, 1994

[54] INCREASING CAPACITY IN WIRELESS BROADCAST SYSTEMS USING DISTRIBUTED TRANSMISSION/DIRECTIONAL RECEPTION (DTDR)

[75] Inventors: Arogyaswami J. Paulraj, Palo Alto; Thomas Kailath, Stanford, both of Calif.

[73] Assignee: The Board of Trustees of the Leland Stanford Junior University, Stanford, Calif.

[21] Appl. No.: 839,624

[22] Filed: Feb. 21, 1992

[51] Int. Cl.[5] H04B 7/06

[52] U.S. Cl. ..... 455/49.1; 455/101; 455/105; 455/27 B; 342/367, 348/384

[58] Field of Search 455/49.1, 51.1, 51.2, 455/59, 60, 61, 101, 103, 105, 137, 273, 278.1; 375/38, 40; 342/367, 463–465; 358/133, 138, 141, 146; 370/59, 63, 64

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4,935,813	6/1990	Fonsalas et al.	358/138
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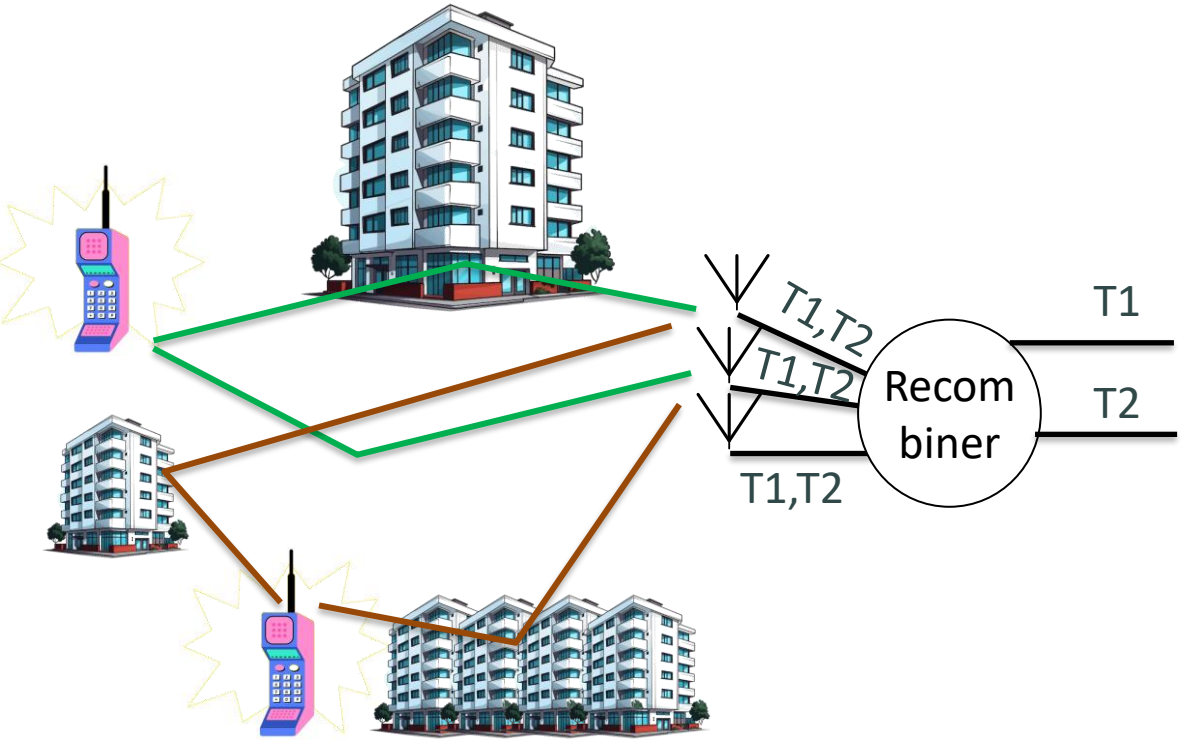
Primary Examiner—Reinhard J. Eisenzopf
Assistant Examiner—Chi Pham
Attorney, Agent, or Firm—Townsend and Townsend Khourie and Crew

[57] ABSTRACT

A method and apparatus for increasing the capacity of wireless broadcast communications system from a central studio to a plurality of users in a service area is disclosed. Given a source signal whose high information rate exceeds the practical information carrying capacity of the available broadcast channel bandwidth, the invention increases the effective capacity of the broadcast system to effectively communicate such a source signal. The high-rate signal is split into several low-rate signals such that each can be accommodated within the allocated bandwidth. These low-rate signals are transmitted from spatially separated transmitters, all radiating into the service area in the same frequency channel. Each receiver uses a plurality of antennas to receive these multiple cochannel signals that arrive from different directions-of-arrival. The receiver exploits the directions-of-arrival differences of these cochannel signals to separate them into the individually transmitted signals. The separated signals are then demodulated to extract the information signals which are then combined to obtain the original high-rate source signal. Thus, the broadcast information capacity can be increased several-fold.

4 Claims, 7 Drawing Sheets

Spatial Multiplexing



T1 reaches all RX antennas

T2 reaches all RX antennas

→ All RX antennas see a superposition of T1 and T2 at same power approximately

→ Undistinguishable

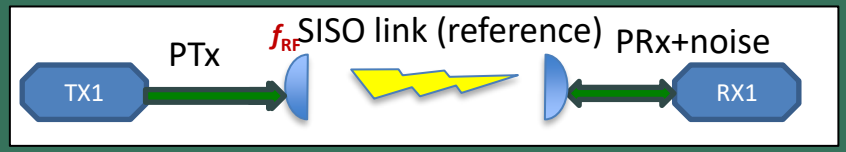
Or…. Are they?

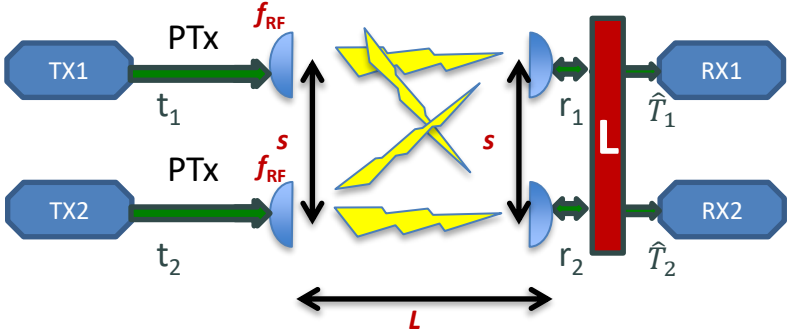
The various time-delayed replicas of T1 reach the RX antennas with different delays

The various time-delayed replicas of T2 reach the RX antennas with different delays

→ The combined signal at the antennas is not “the same”, they see subtle differences in delays

Simple Version of 2x2 MIMO





$$r_1(t)=t_1(t-\tau_{direct})+t_2(t-\tau_{cross})$$

$$r_2(t)=t_2(t-\tau_{direct})+t_1(t-\tau_{cross})$$

τ_{direct}: propagation delay between 1-1 and 2-2, $\propto L$

τ_{cross}: propagation delay between 1-2 and 2-1, $\propto \sqrt{(L^2+s^2)}$

$$r_1(t+\tau_{direct})= t_1(t)+t_2(t+\tau_{direct}-\tau_{cross})$$

$$r_2(t+\tau_{direct})=t_2(t)+t_1(t+\tau_{direct}-\tau_{cross})$$

t1 and t2 are modulated carriers, e.g. $t_1=\mathrm{Re}\{T_1(t)\, e^{j\omega_{RF}t}\}$; $t_2=\mathrm{Re}\{T_2(t)\, e^{j\omega_{RF}t}\}$

Small delays: at $v=3\cdot10^8$m/s , $\tau_{direct}-\tau_{cross}$ is very small, $\propto \sqrt{(L^2+s^2)}-L\approx L(1+s^2/(2L^2)-1)=s^2/(2L)$ → a few nsec

It is a delay relevant for the carrier frequency (tens of GHz) but not for symbol duration (tens of Msym/s) → Cplx env.

$R_1=T_1+T_2e^{-j\Delta\phi}$

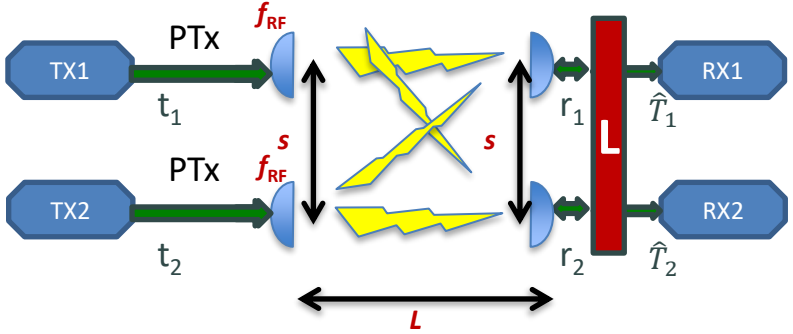
$R_2=T_2+T_1e^{-j\Delta\phi}$

$$\begin{bmatrix}R_1\\R_2\end{bmatrix}=\begin{bmatrix}1 & e^{-j\Delta\varphi}\\ e^{-j\Delta\varphi} & 1\end{bmatrix}\begin{bmatrix}T_1\\T_2\end{bmatrix} \rightarrow \begin{bmatrix}\hat{T}_1\\\hat{T}_2\end{bmatrix}=\begin{bmatrix}1 & e^{-j\Delta\varphi}\\ e^{-j\Delta\varphi} & 1\end{bmatrix}^{-1}\begin{bmatrix}R_1\\R_2\end{bmatrix}=\begin{bmatrix}T_1\\T_2\end{bmatrix}$$

L

Transmitted signals always retrievable?

Simple Version of 2x2 MIMO



$$\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} 1 & e^{-j\Delta\varphi} \\ e^{-j\Delta\varphi} & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

There is additive noise!

$$\begin{bmatrix} \hat{T}_1 \\ \hat{T}_2 \end{bmatrix} = \begin{bmatrix} 1 & e^{-j\Delta\varphi} \\ e^{-j\Delta\varphi} & 1 \end{bmatrix}^{-1} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \begin{bmatrix} 1 & e^{-j\Delta\varphi} \\ e^{-j\Delta\varphi} & 1 \end{bmatrix}^{-1} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

$$= \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \frac{\begin{bmatrix} 1 & -e^{-j\Delta\varphi} \\ -e^{-j\Delta\varphi} & 1 \end{bmatrix}}{1-e^{-2j\Delta\phi}} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \frac{\begin{bmatrix} 1 & -e^{-j\Delta\varphi} \\ -e^{-j\Delta\varphi} & 1 \end{bmatrix}}{e^{-j\Delta\phi} 2j \sin\Delta\phi} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

When $\Delta\phi \to 0$, the channel matrix is no longer invertible, e.g. noise explodes, much bigger than desired T1 and T2!

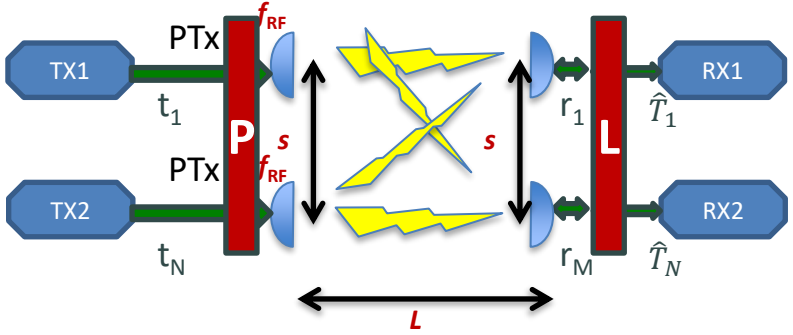
Could you expect it?

When $\Delta\phi=\pi/2 \to \begin{bmatrix} \hat{T}_1 \\ \hat{T}_2 \end{bmatrix} = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \frac{\begin{bmatrix} 1 & j \\ j & 1 \end{bmatrix}}{e^{-j\Delta\phi} 2j} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \to \hat{T}_1 = T_1 + n_1/2 + n_2/2 \to SNR_{MIMO}=P_{Rx}/(Power(n_1)/4+Power(n_2)/4)=2\,SNR_{SISO}$

This is "Zero Forcing" recombination. Other approaches available (e.g. Maximum Ratio), but all subject to same constraint:

depending on Δϕ ("diversity"), the original streams can be recovered (even improved) or not (noise enhancement)

More Generally



$$\begin{bmatrix} R_1 \\ R_N \end{bmatrix} = \boldsymbol{M} \begin{bmatrix} T_1 \\ T_M \end{bmatrix} + \begin{bmatrix} n_1 \\ n_N \end{bmatrix}$$

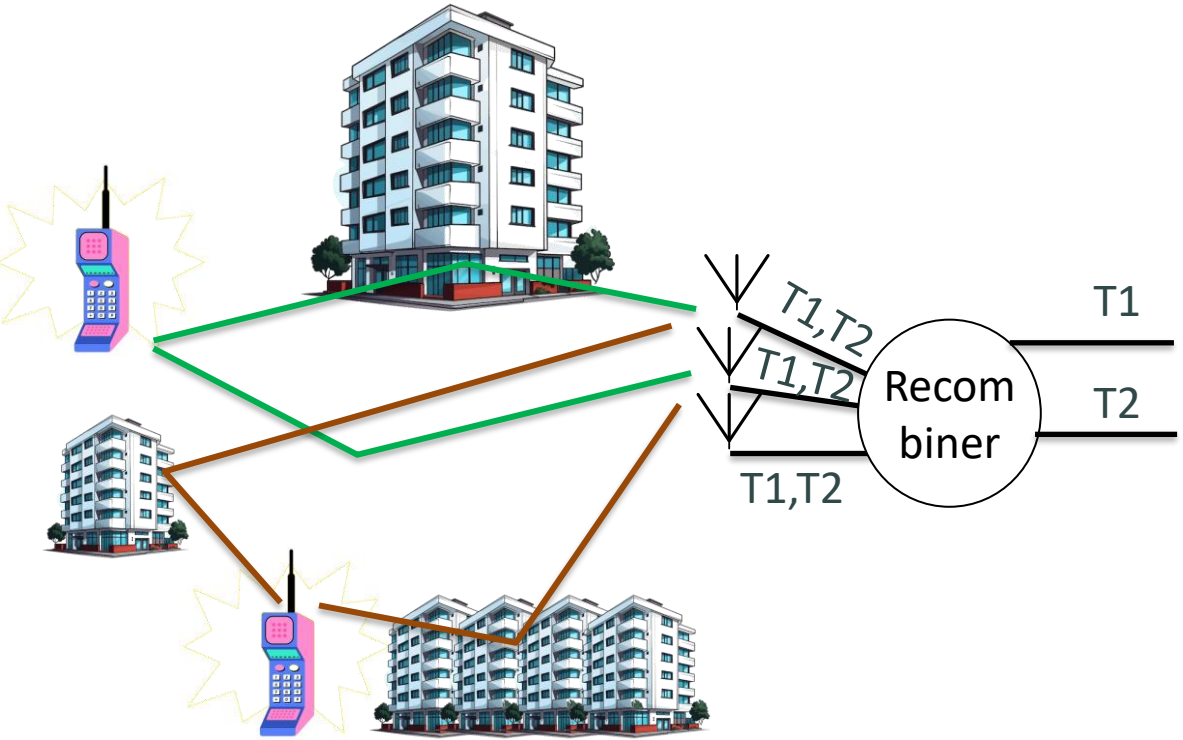
Optimum:

$\boldsymbol{M} = \boldsymbol{U}\,\boldsymbol{\Lambda}\,\mathbf{V}^{\mathbf{T}}$ (SVD decomposition)

Number of non-zero singular values
≈number of parallel channel attainable
U: describes optimum precoding (in block P, at Tx side)
V: describes optimum postprocessing (in block L, at Rx side)

Precoding: applicable only if the transmitters are all in the same place
Synchronization: ideally necessary; true performance drops if all Rx and Tx use different clocks/timing/carrier
Number of antennas: can be different among TX and RX sides (but max number of parallel channels ≤min(M,N))

MIMO: Requirements



Multiple antennas at TX

Multiple antennas at RX

Diversity

- In LOS systems: physical spacing of antennas [point-to-point high-speed links]
  →Tx and Rx must be in the "near field" region of the overall antenna system (but in the far field of the basic antennas)
- In non-LOS: environment reflections [home router, mobile communications]

A lot of hardware and software

MIMO Hardware

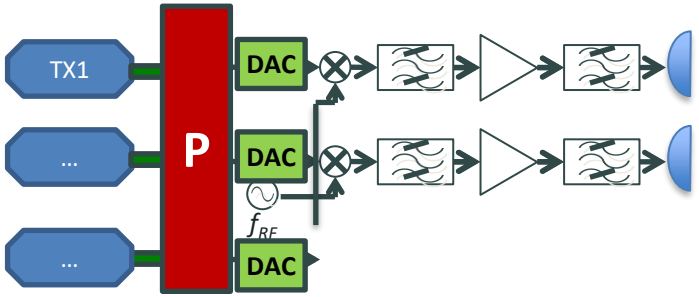
MIMO Transmitter:

Several digital streams, synchronized

Digital preprocessing **P**

Nx DAC, Nx upconversion chains, Nx antennas

Local oscillator must be split and distributed to all (or at least common precise reference clock for synthesized generators)



P precombines the streams → Signals in chains are combinations of modulated carriers → High PAPR! → Non lin. dist.

MIMO Receiver:

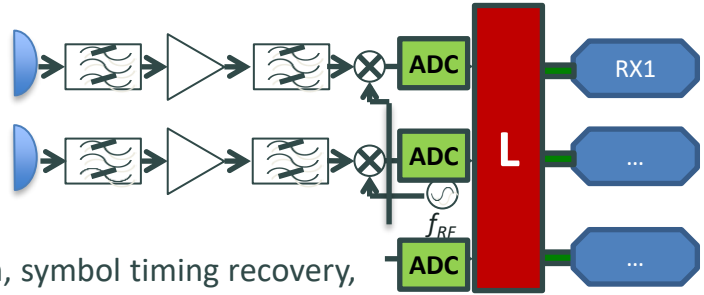
Several analog signals with high PAPR, entirely mixed

Digital postprocessing L

Nx antennas, Nx downconversion chains, Nx ADC

Local oscillator distribution

Digital processing is very heavy (at N x symbol rate: carrier estimation, symbol timing recovery, extraction of data, estimation of channel matrix, adjustment of MIMO cancelling coefficients)



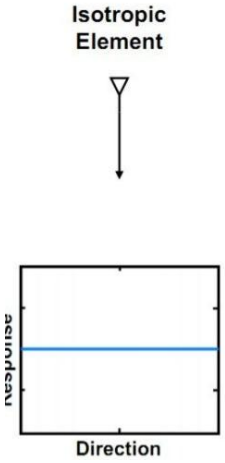
Agenda

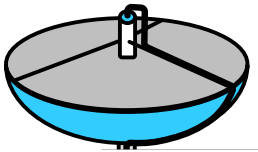
- MIMO
- Phased Arrays
- Multi-carrier
- Wrap-up: Architectures and Components

Phased Array

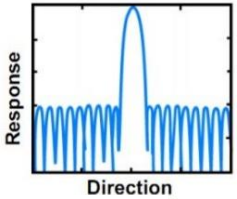
Several Antennas, fed by a common Tx Signal

Basic example: Isotropic antenna vs many isotropic antennas



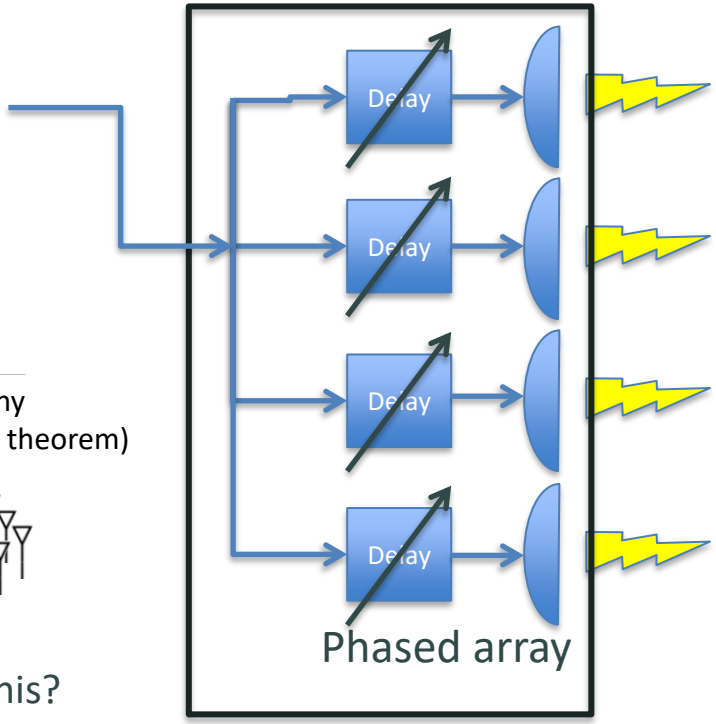


A large aperture is equivalent to many isotropic antennas in it (equivalence theorem)





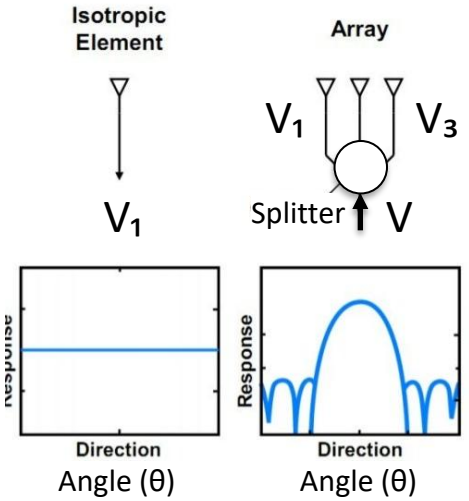
How to verify this?

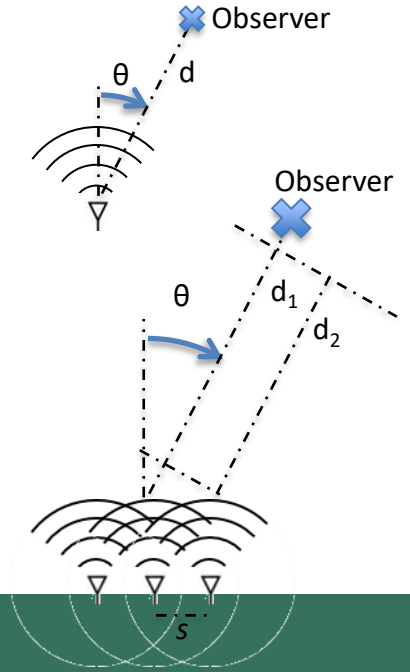


Phased Array

Several Antennas, fed by a common Tx Signal

Basic example: Isotropic antenna vs many isotropic antennas





Wave number or phase constant: $\beta=2\pi/\lambda$

Each antenna radiates a spherical wave $E_i \propto V_i e^{-j\beta d}/d$

Feeding all antennas with the same signal $V_i = V/\sqrt{N}$

All spaced at spacing *s*

In far field, all waves add:

$$R \propto E_{tot} \propto \Sigma V_i \, e^{-j\beta d_i}/d_i$$

Each with its own distance to the observer (all approximately parallel and $d \approx d_i$ at denom):

$d_i = d + i\, s \sin\theta$

$$R \propto \frac{V}{\sqrt{N}} e^{-j\beta d} \frac{\sum_i e^{-ji\beta s \sin\theta}}{d}$$

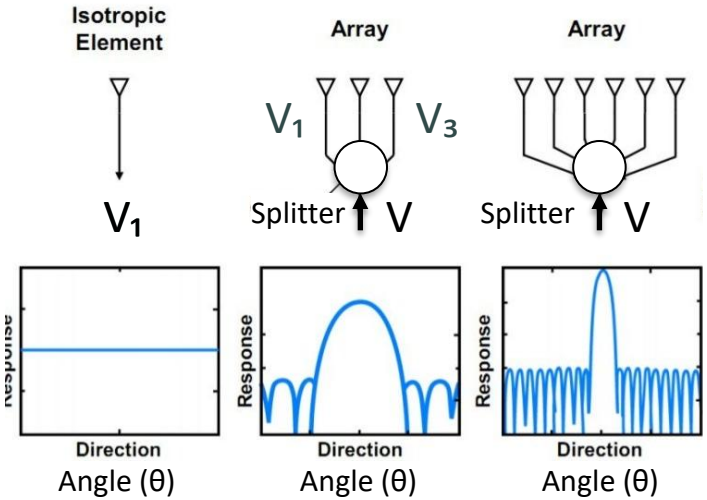
When θ=0 all waves are in phase!

Moving away from θ=0: contributions no longer in phase → Directive antenna!

Phased Array

Several Antennas, fed by a common Tx Signal

Basic example: Isotropic antenna vs many isotropic antennas



Wave number or phase constant: $\beta=2\pi/\lambda$

Each antenna radiates a spherical wave $E_i \propto V_i e^{-j\beta d}/d$

Feeding all antennas with the same signal $V_i = V/\sqrt{N}$

All spaced at spacing *s*

Directivity increases as more antennas are added, fed in phase:

$$R \propto \sum_i \frac{V_i e^{-ji\beta d_i}}{d_i} = \frac{V}{\sqrt{N}} e^{-j\beta d} \frac{\sum_i e^{-ji\beta s \sin\theta}}{d}$$

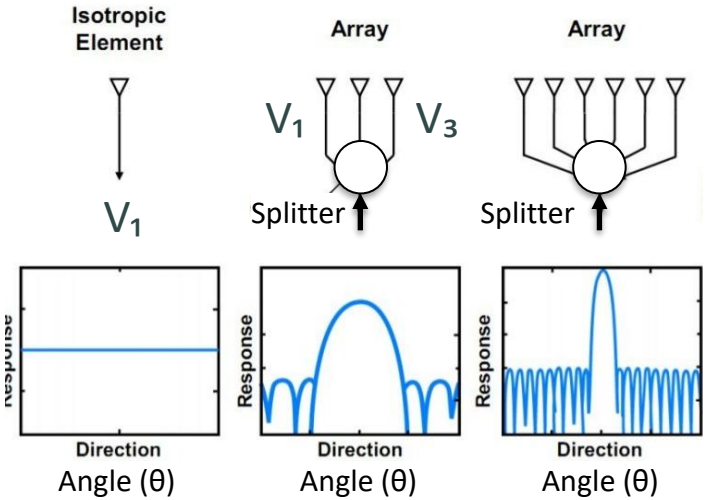
Moving away from θ=0, quickly the N contributions no longer combine in phase

→ Narrower beam → Higher directivity

Phased Array

Several Antennas, fed by a common Tx Signal

Basic example: Isotropic antenna vs many isotropic antennas



Wave number or phase constant: $\beta=2\pi/\lambda$

Each antenna radiates a spherical wave $E_i \propto V_i e^{-j\beta d}/d$

Adding a progressive phase via a phase shifter:

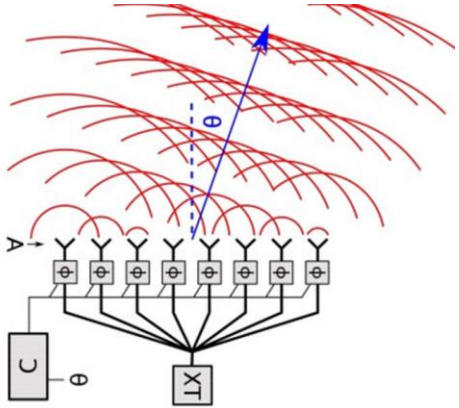
$V_i=V/\sqrt{(N)}\, e^{j\, i\Delta\varphi}$

$$R \propto \sum_i \frac{V_i e^{-ji\beta d_i}}{d_i} = \frac{V}{\sqrt{N}} e^{-j\beta d} \frac{\sum_i e^{-ji(\beta s \sin\theta - \Delta\phi)}}{d}$$

All waves combine constructively for θ such that:

$\theta=\beta s \sin\theta - \Delta\varphi$

→ $\theta=\arcsin(\Delta\varphi/(\beta s))$



We can steer the beam by changing phases!

Phased Array Pattern and Feeding (or Illumination)

Can emulate any antenna, including high-directivity ones!

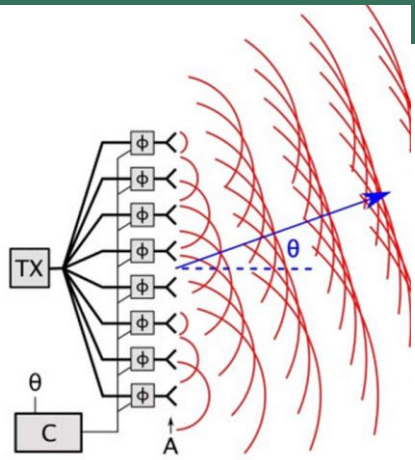
Final pattern: "Array pattern" (with isotropic elements) x rad. pattern of antenna element

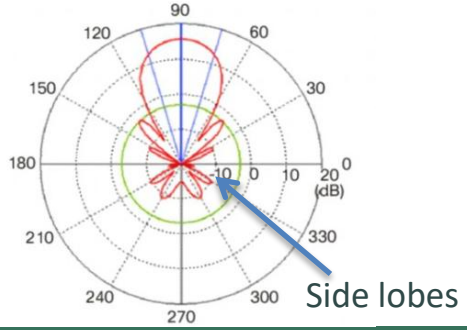
Gain: $GainArray_{dBi} = 10\text{Log}(NumAntennas) + GainOfBaseElement_{dBi}$ (e.g. patch antenna $\approx 5dBi$)

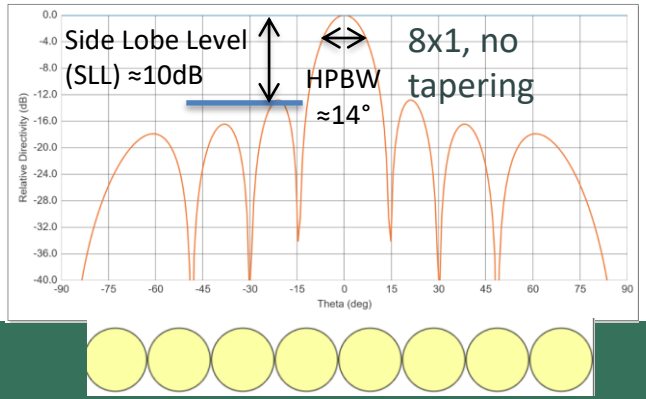
Phases of elements → Beam steering (maximum radiation direction can be changed)

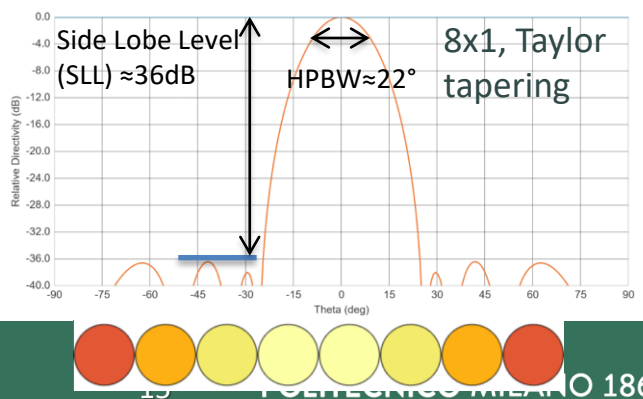
Amplitudes: adding attenuators or amplifiers on each antenna

- Uniform (all with same power): high secondary lobes
- Non uniform (e.g. tapering): elements at edges have lower power
  → Lower secondary lobes, but less overall power and directivity (smaller eff. area)









Phased Arrays: Grating Lobes

Recombination of all elements in phase:

$$R \propto \sum_i \frac{V_i e^{-ji\beta d_i}}{d_i} = \frac{V}{\sqrt{N}} e^{-j\beta d} \frac{\sum_i e^{-ji(\beta s \sin\theta - \Delta\phi)}}{d}$$

A few slides earlier: "Waves combine constructively: $\theta=\beta s \sin\theta-\Delta\varphi \rightarrow \theta=arcsin(\Delta\varphi/(\beta s))$"

Rather, more generally:

max recombination at θ such that: $2N\pi=\beta s \sin\theta-\Delta\varphi \rightarrow \theta=arcsin\left((\Delta\varphi+2N\pi)/(\beta s)\right)$

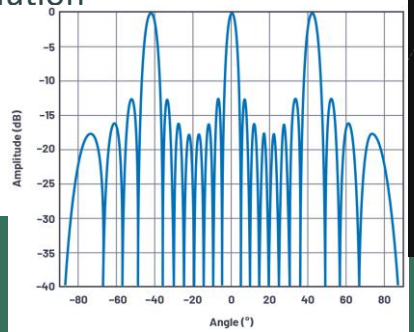
There can be MANY directions of maximum:

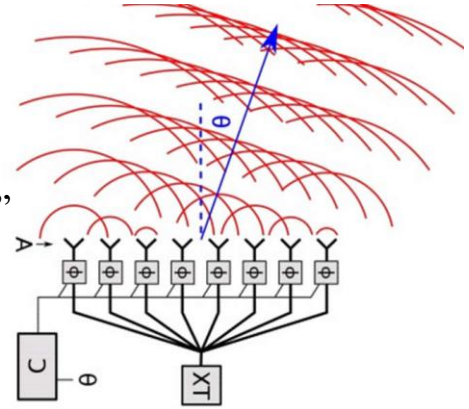
- For s≤λ/2 : βs<π→ $\theta=arcsin(\Delta\varphi/\pi+2N)$
  →arcsin gives only one solution within ±π/2

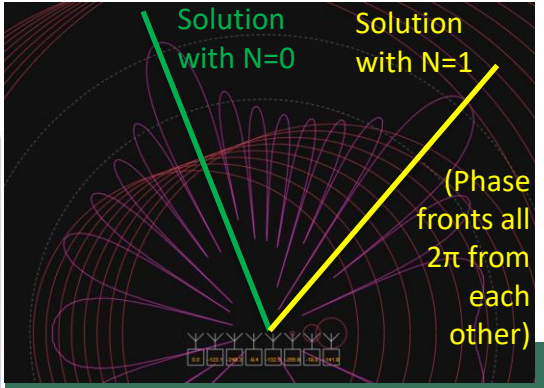
"Normal" spacing for a phased array: s=λ/2

- For s>λ: → arcsin will always give more than one solution

Grating lobes are normally undesired
(passive interference, active interference, uncertainty in detection of direction-of-arrival, radiation in forbidden directions…)







Phased Array: Play!

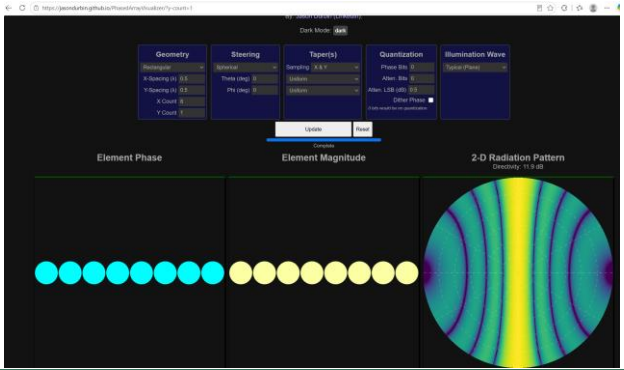
Justify all previous phenomena:

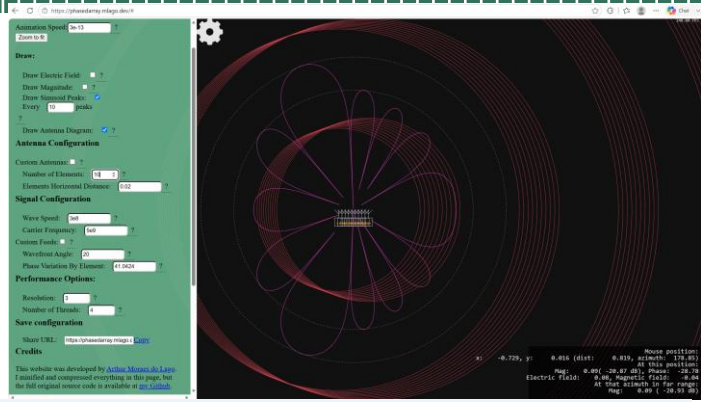
Radiation pattern is the Fourier Transform of the illumination
→ Aliasing, truncation/tapering, beam steering

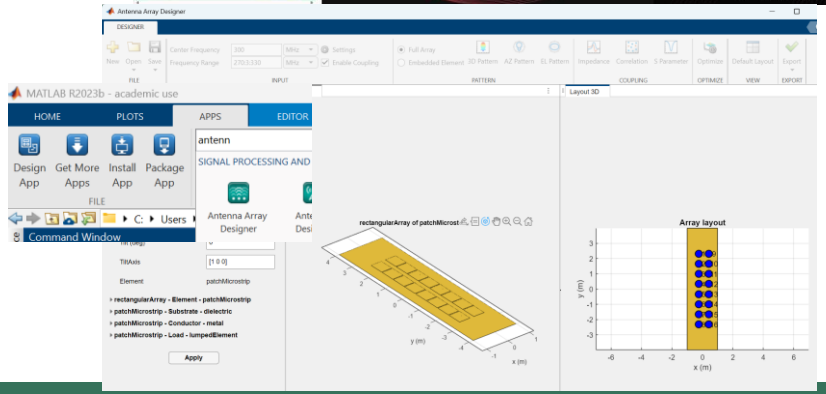
Used to also place “nulls” in desired directions (against interferers)

Play with available tools, e.g.:

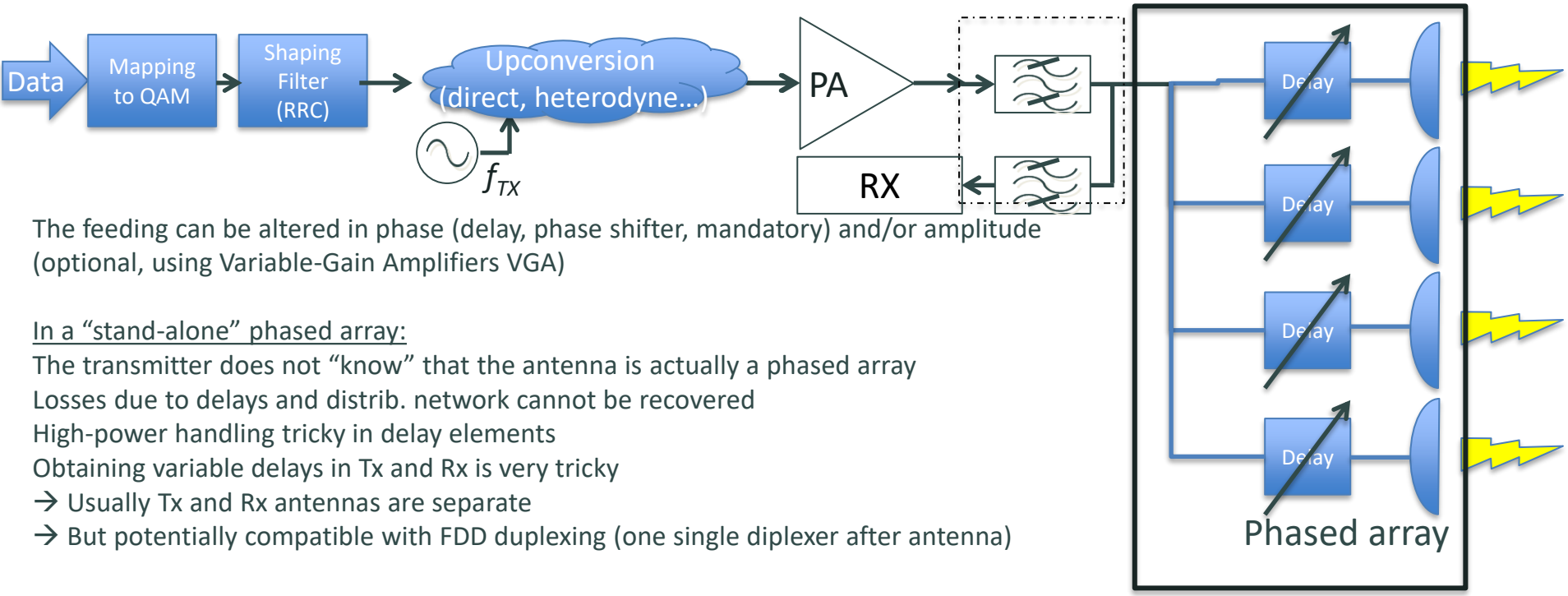
- https://phasedarray.mlago.dev/#
- Matlab Antenna Array Designer
- https://jasondurbin.github.io/PhasedArrayVisualizer







Phased Array Architecture: Stand-alone Antenna



The feeding can be altered in phase (delay, phase shifter, mandatory) and/or amplitude (optional, using Variable-Gain Amplifiers VGA)

<u>In a “stand-alone” phased array:</u>

The transmitter does not “know” that the antenna is actually a phased array

Losses due to delays and distrib. network cannot be recovered

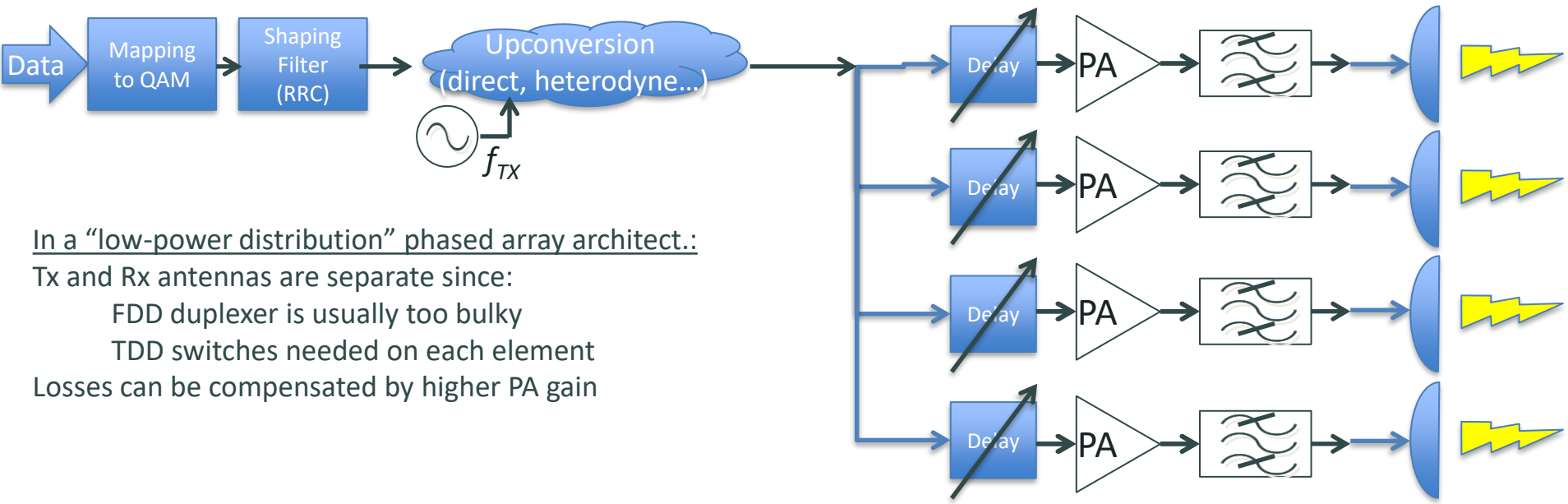
High-power handling tricky in delay elements

Obtaining variable delays in Tx and Rx is very tricky

→ Usually Tx and Rx antennas are separate

→ But potentially compatible with FDD duplexing (one single diplexer after antenna)

Phased Array Architecture: Low-power distribution



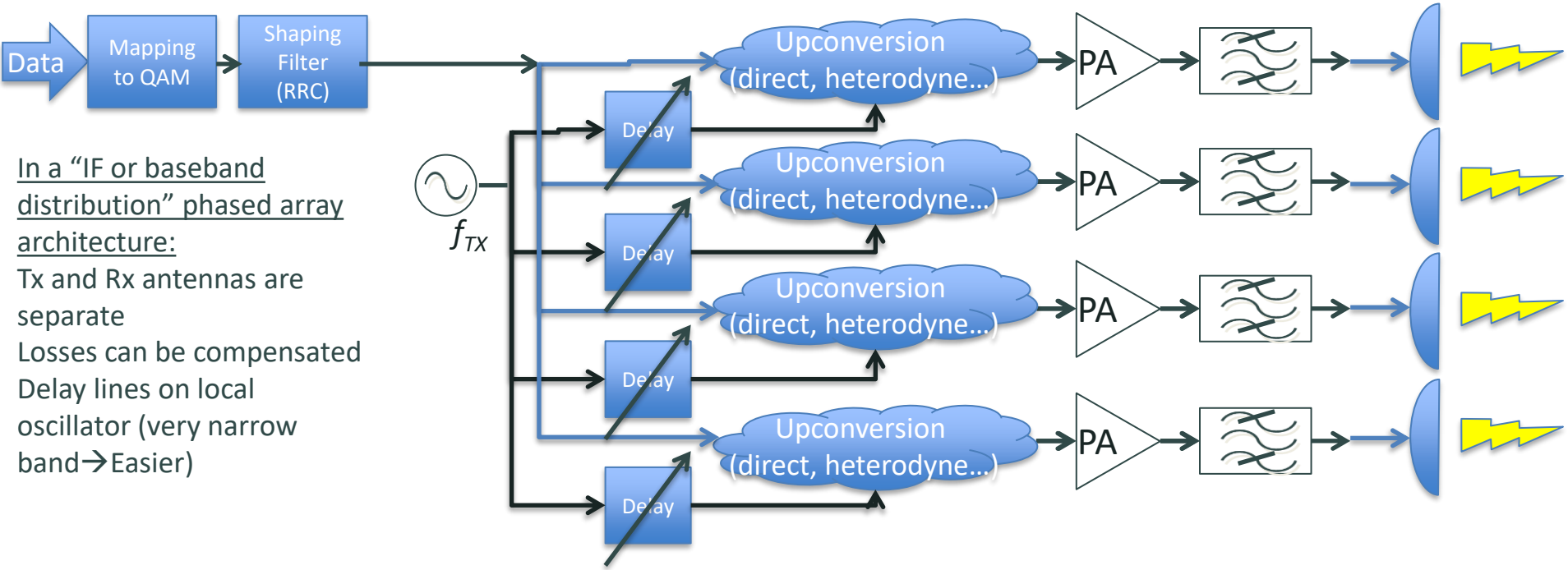
In a “low-power distribution” phased array architect.:

Tx and Rx antennas are separate since:

- FDD duplexer is usually too bulky
- TDD switches needed on each element

Losses can be compensated by higher PA gain

Phased Array Architecture: IF or baseband distribution



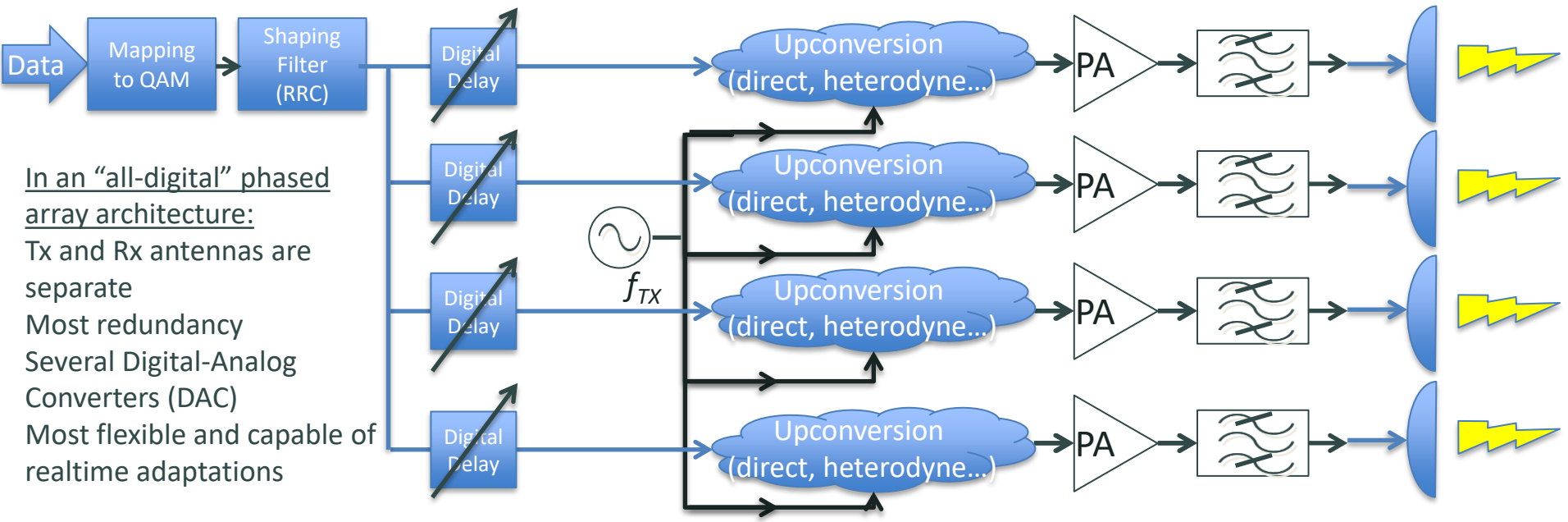
In a "IF or baseband distribution" phased array architecture:

Tx and Rx antennas are separate

Losses can be compensated

Delay lines on local oscillator (very narrow band→Easier)

Phased Array Architecture: All Digital



In an "all-digital" phased array architecture:

- Tx and Rx antennas are separate
- Most redundancy
- Several Digital-Analog Converters (DAC)
- Most flexible and capable of realtime adaptations

All-digital architecture is compatible with MIMO systems

Agenda

- MIMO
- Phased Arrays
- Multi-carrier
- Wrap-up: Architectures and Components

Multi-Carrier (OFDM)

Given a carrier frequency and a bandwidth *B*

B is split in *N* sub-channels, each *B/N* wide, but not RRC-filtered → side lobes overlap

Each sub-channel is treated as a low-capacity channel, with its own symbols and sub-carrier

A symbol *S* in subcarrier *k* lasts *N/B* in time: (complex envelope, later upconverted to RF):

$$V(t)=\Sigma_k\, S_k\, e^{j\,\Omega_k t}$$

Sampling time discretely at Δt=1/B: $V_i=\Sigma_k\, S_k\, e^{j\,\Omega_k i\,\Delta t}$

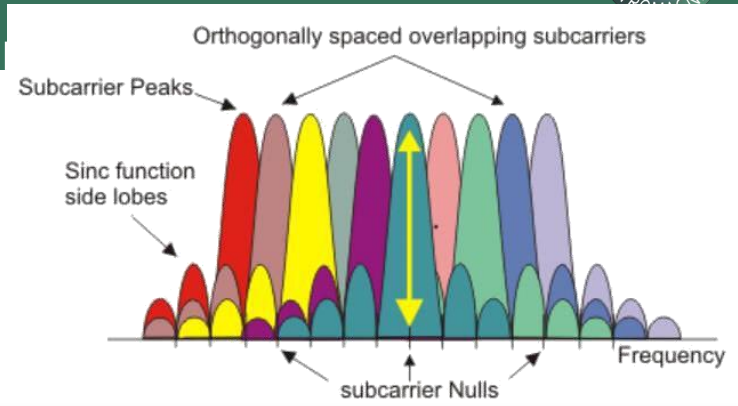
If Ω_k=k2π/(NΔt):

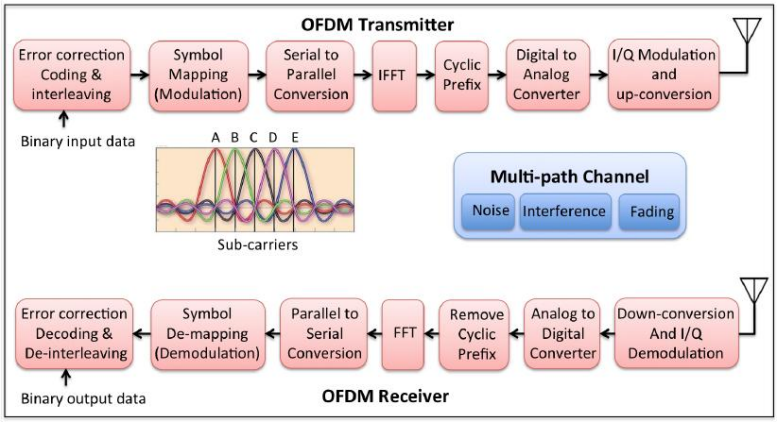
$$V_i=\Sigma_k S_k\, e^{j\,2\pi\, k\, i/N} \rightarrow \underline{\boldsymbol{V}}=\mathrm{IFFT}(\underline{\boldsymbol{S}})$$

Given a block of *N* symbols $\underline{\boldsymbol{S}}$, we obtain a segment of complex waveform in time $\underline{\boldsymbol{V}}$, ready for upconversion & transmission!

In RX: $\underline{\boldsymbol{S}}=\mathrm{FFT}(\underline{\boldsymbol{V}})$

Subcarriers are intrinsically orthogonal (although overlapping)!





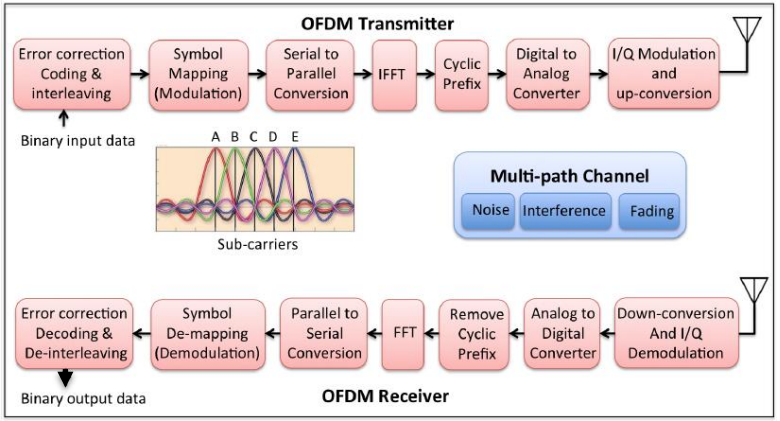
OFDM Architecture

- Robust to time-spreading & echoes thanks to guard intervals "pauses" between symbols
- Robust against frequency-selective fading:

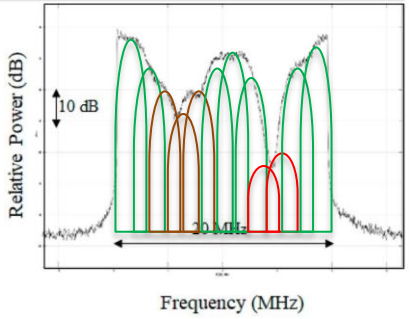
  If some frequencies are attenuated (e.g. multipath), some subcarriers will not survive → The others survive!
- Modern OFDM: WiFi, 4G, 5G, WiMax, DAB, DVB-T, CableTV DVBC
- Flexible allocation of resources in multi-user scenario: e.g. in 5G a "Resource block" covers 12 subcarriers for some symbols

Architecture Implications:

- Lots of Digital Signal Processing and dedicated ASIC
- High PAPR ≈10Log10(N) (e.g. 32k FFT)→ PAPR=45dB (statistically +"crest factor" reduction→PAPR≈10÷15dB)



Needs linearization! (e.g. predistortion)



Agenda

- MIMO
- Phased Arrays
- Multi-carrier
- Wrap-up: Architectures and Components

Modern Systems Architectures

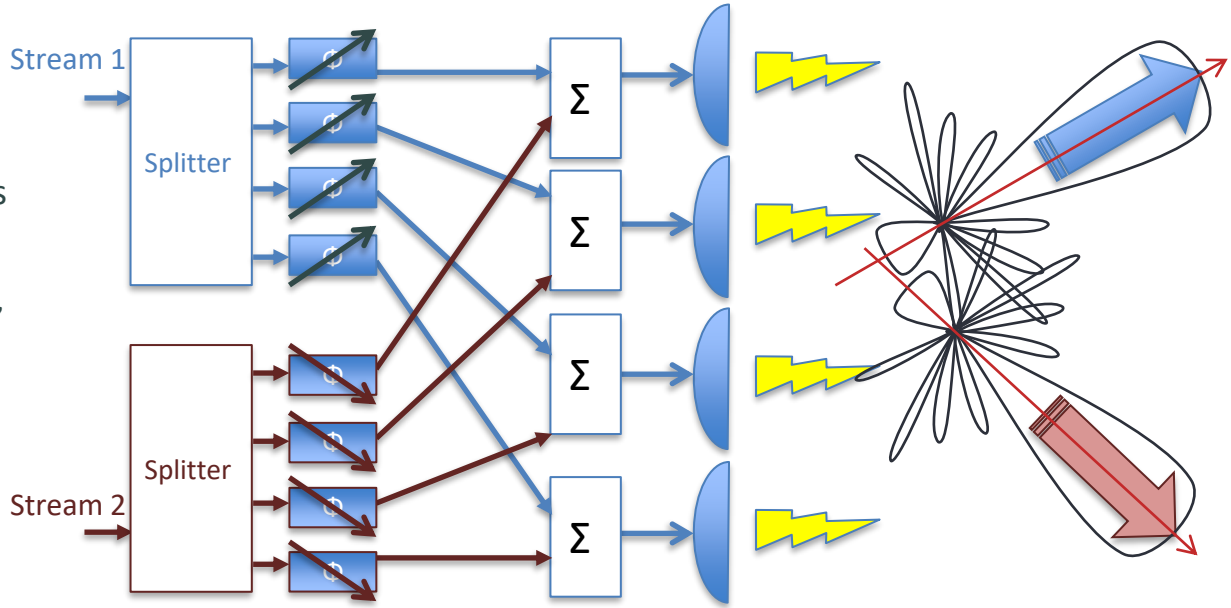
Coexistence of analog and digital

Phased Arrays:

- Electronic steering
- Dynamic tracking of users/targets
- Null placement
- Similar size as directive antennas, but with higher losses due to distribution/variable phases…
- Multi-beam Phased Array

MIMO for enhanced capacity or coverage or tracking

Multi-user with flexible allocation of resources



More Components

Variable-Gain Amplifiers

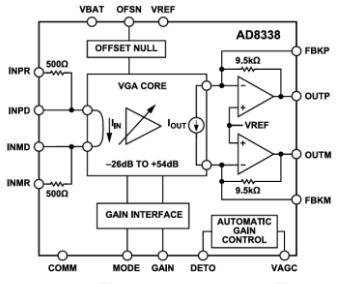
- With internal digital attenuators
- Changing the actual gain of amplifier

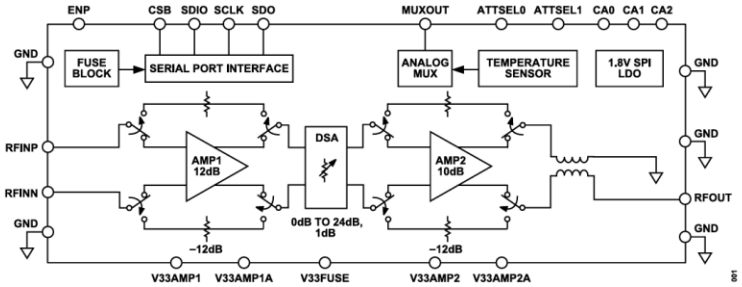
Variable Attenuators

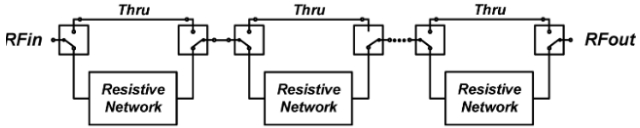
- Digital step attenuators

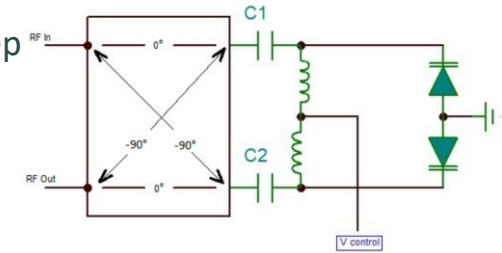
Phase Shifters

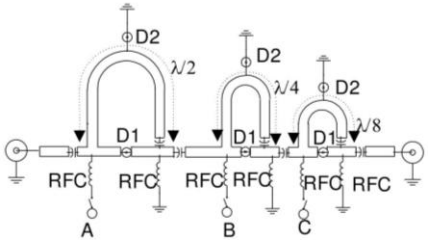
- Fixed: delay elements (transmission line)
- Variable in digital step
- Variable analog











Combiners, Splitters, Couplers, Hybrids (second part of the course)

